Do Now · retrieval

Thinking concurrently: independent tasks

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does the MAR contain during a memory access?

Q2

When is A AND B true?

Q3

What distinguishes iterative development from a fixed sequence?

Q4

Search for 13 in sorted indexes 0–14. Trace the midpoint indexes inspected by standard integer midpoint selection.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Find tasks without dependencies — Dataset download and image processing can operate concurrently if they use separate resources.

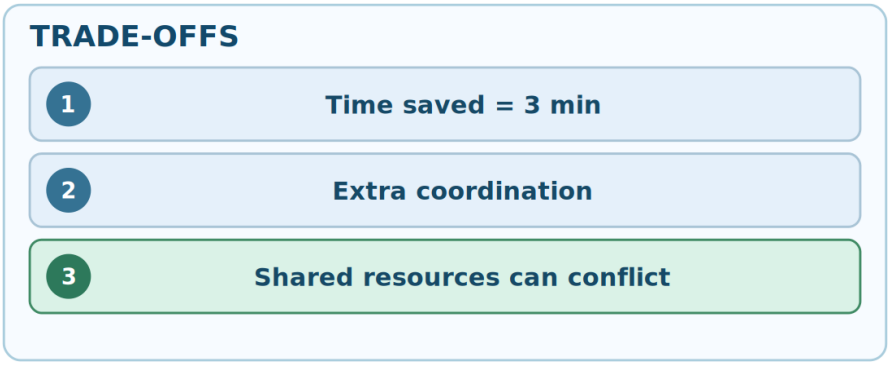
Core idea 2

Respect the final dependency — Report assembly cannot start until BOTH preparation tasks finish.

Core idea 3

Calculate the ideal gain — Sequential 9 minutes, parallel 6 minutes. Actual performance depends on scheduling and resource contention.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

concurrency — Tasks making progress over overlapping intervals.

dependency — A task needs another result first.

critical path — Longest dependent chain determining ideal completion time.

race condition — Timing-dependent outcome caused by unsafe shared access.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Two independent data checks take 5 and 3 minutes; publishing takes 2 more. Calculate ideal sequential and concurrent times.

Follow the worked method

1. Predict the method — Add all durations for the sequential path: 10 minutes.

2. Check and explain — Run independent checks together, wait 5 minutes, then publish in 2: total 7 minutes.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict the method
Images and CSS are independent preparation tasks; calculate 4+6+1=11 minutes sequentially.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Tasks A (3 min) and B (5 min) are independent, then final C (2 min). Find sequential and ideal parallel completion times.

Question 2

Two withdrawals of £10 and £20 both read the same initial £100 balance before either writes an update. Explain a possible lost update and propose a safeguard.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Four tasks: parse A=4 min, fetch B=6 min, validate C=2 min (requires A and B), publish D=1 min (requires C). Find shortest ideal time and explain two constraints.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

All parts of an algorithm can always run concurrently if enough cores are available.

Use key vocabulary from memory

1. Explain concurrency in your own words.

2. Explain dependency in your own words.

3. Explain critical path in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.